

Web Robots: The Worker Bees of the Internet

Web robots, also known as Web crawlers and Web spiders, traverse the Internet to extract various types of information. Web robots can be used for legitimate as well as malicious purposes. Read more about them and make a simple robot of your own by following the directions given in this article.



When we come across the term 'robot', most of us think about the hardware robots that are replacing humans in doing various tasks at an alarming rate. Some of us may even have this vision of robots playing badminton, serving food, etc. According to software entrepreneur and futurist, Martin Ford, it is estimated that around 75 per cent of the work carried out by humans nowadays will be taken over by robots or machines by the end of this century. I am sure many of us have probably seen all that robots can do in *Terminator* and other such Hollywood movies. So if robots can carry out all such physical activities, then can a robot click on a Web button, Web check box or enter different input values in an input box? Certainly, it can perform all such tasks as well, and we call such robots Internet bots or Web robots.

A Web robot is simply a software application (or a set of programs) that traverses any Web application and performs various automated tasks. Though these robots can perform various tasks over a Web application, they are widely used to perform repetitive tasks, which they do much faster than if performed manually. Web robots, also referred to as Web spiders or Web crawlers, are nothing but scripts which

retrieve information from different Web servers, analyse them and then file them at a speed that is many times faster than the rate humans would perform the same task. We should not be surprised to learn that more than half (52 per cent, according to Imperva) of all the Web traffic is actually enabled by Web robots, as most visitors to websites are not humans but Web bots. They are referred to as the worker bees of the Internet. Some of the Web bots help in refreshing your Facebook feeds continuously whereas some, like Googlebots, help us in figuring out how to rank different Google search results. Web bots make use of scripts to interact with different Web applications. Hence, we can develop Web robots using different open source scripting or programming languages like Perl, Python, C, PHP, etc.

All of these languages allow Web bot developers to write scripts that can actually perform different procedural tasks such as Web scanning, Web indexing, etc. Web bots can be made to perform many other different types of tasks with the help of these scripts and, hence, they are categorised based on the kind of actions performed by them on any Web application. There are malicious Web robots as well, which are used by hackers to perform diverse tasks on the Internet by installing some